



Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences
Department of Computer Science & Information Technology

Course Code with Title	CS 325T: Artificial Intelligence		Program	BS (CS)
Instructor	Muhammad Muneeb		Semester	6th
Start date & Time	6 th May, 2025 at 3:30 PM	End Time	5:00 PM	
Maximum Marks	30			

IMPORTANT INSTRUCTIONS:

Read the following Instructions carefully:

- Attempt All Questions.
- Calculator Are Allowed.

Q.No.1	(CLO_1): (Cognitive Level C2, i.e., Understanding Problems)	(PLO_2: Knowledge for Solving Computing (04+04+04+04+04 Marks))
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(a) **Differentiate** between the following:

- 1: Uninformed Search vs. Heuristically Guided (Informed) Search
- 2: General Artificial Intelligence vs Narrow Artificial Intelligence
- 3: Single-Agent Systems vs. Multiagent Distributed Systems
- 4: Model-Based Deliberative Agent vs. Simple Reflexive Reactive Agent

Ans: 1 (A) part

Uninformed Search	Informed Search
1: Uses only problem definition; no additional information.	Uses problem definition plus heuristic knowledge to guide search.
2: Usually slower, explores more nodes blindly.	More efficiently, explore fewer nodes by using heuristics.
3: Breadth-First Search, Depth-First Search.	A* Search, Best-First Search, Greedy Search.

General Artificial Intelligence	Narrow Artificial Intelligence
1: Can perform any intellectual task like a human.	Specialized for specific tasks only.
2: Adapts to unfamiliar tasks and learns autonomously.	Performs only predefined tasks, limited adaptability.
3: Still theoretical, under research.	Widely deployed (e.g., chatbots, recommendation systems)



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Single Agent Systems	Multi Agent Distributed System
1: One agent operating in the environment.	Multiple agents interacting and working together.
2: No need for coordination with others.	Requires coordination, cooperation, or negotiation.
3: Generally simpler design and control.	More complex due to interactions among agents.

Model-Based Deliberative Agent	Simple Reflexive Reactive Agent
1: Uses an internal model of the world to make decisions.	Responds directly to stimuli without internal model.
2: Can plan and anticipate future states	Acts only based on current perceptual input.
3: More complex, slower but flexible	Simple, fast but limited in capability.

(b) **Discuss** for each of the following agents Performance, Environment, Actuators and Sensors (PEAS):

- 1: Automated Taxi Driver
- 2: Chat Bots
- 3: Medical Diagnosis
- 4: Image Analysis

(B) PART

AGENTS	PERFORMANCES	ENVIRONMENT	ACTUATORS	SENSORS
Automated Taxi Driver	➤ Safe driving ➤ Quick arrival ➤ Passenger comfort	➤ Roads ➤ Traffic ➤ Pedestrians	➤ Steering ➤ Accelerator & brake ➤ Indicators/horn	➤ Cameras ➤ GPS ➤ Radar
Chat Bots	➤ Relevant answers ➤ Fast response ➤ User satisfaction	➤ Text ➤ Conversations ➤ User queries	➤ Text generation ➤ Send Messages ➤ Display updates	➤ Text input ➤ Speech-to-text ➤ Context detection
Medical Diagnosis	➤ Accurate diagnosis ➤ Timely results ➤ Patient safety	➤ Hospitals ➤ Patient records ➤ Lab results	➤ Report generation ➤ Alerts ➤ Recommendation systems	➤ Patient data ➤ Medical tests ➤ Symptom input
Image Analysis	➤ Correct identification ➤ High accuracy ➤ Fast processing	➤ Digital images ➤ Databases ➤ Camera feeds	➤ Processed outputs ➤ Tags/classifications ➤ Alerts	➤ Cameras ➤ Image files ➤ Scanners



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- (c) **Describe** the Autonomous Agent scenario of a robot in a big library and wants to find a specific book. The robot does not know the exact location of the book. It needs to plan the path and search the book from different shelves. The robot must avoid obstacles like tables and chairs. How will the robot act as a problem-solving agent?

(C) PART

In the autonomous agent scenario, the robot acts as a problem-solving agent first defining its goal of finding a specific book in the large library. Since it does not know the exact location, it uses sensors to perceive its environment, building a map of shelves, tables, and chairs. It plans a path by breaking the problem into smaller tasks, such as searching one shelf at a time. The robot applies a search strategy, like depth-first or breadth-first search, to systematically check each shelf. As it moves, it updates its map and avoids obstacles using obstacle-detection sensors. If it encounters a blocked path, it replans its route dynamically. The robot uses feedback from its sensors to adjust its movement and keep track of visiting areas. Once it finds the book, it stops and possibly informs the user. Throughout, the robot balances exploration and efficiency to achieve its goal effectively.

- (d) **Explain** the Estimated Cost and Shortest Path by discovering the paths from starting node 'S' to goal node 'G' as shown in Fig: 01 by using **A* Algorithm** of informed search.

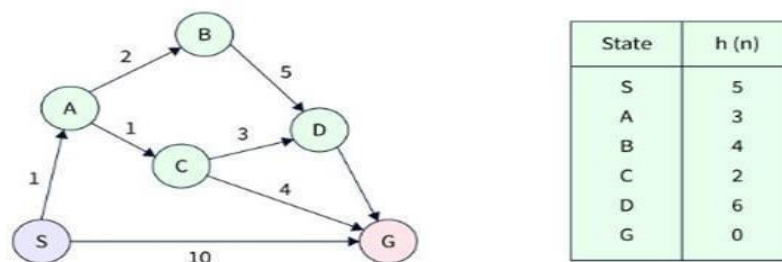
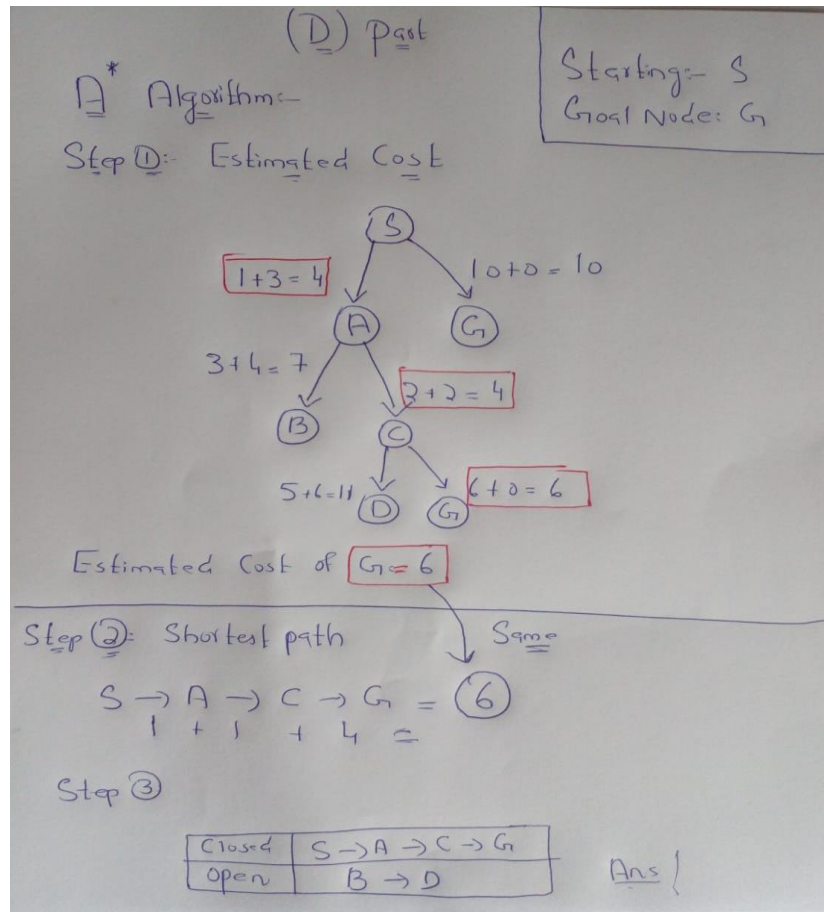


Fig: 01



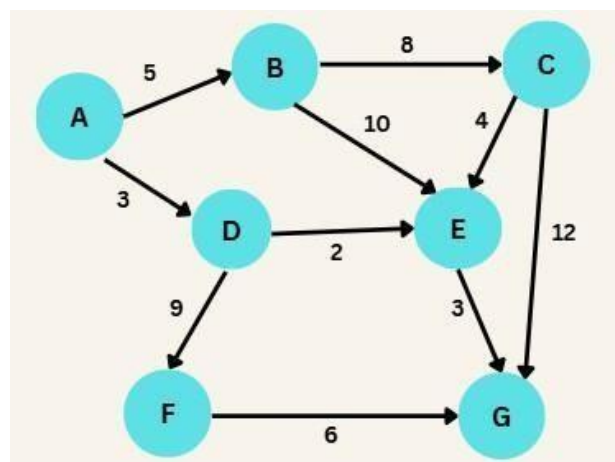
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- (e) **Consider** the Given Graph, and **Illustrate** the reached goal, minimum path cost using Uniform Cost Search (UCS)

Starting State: A

Goal State: G





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(E) Part

Uniform Cost Search

Starting state: A
 Goal state: G

Step ①:

Open list	Closed list
① A(0)	[]
② AB(5), AD(3)	A(0)
③ AB(5), DE(2), DF(9)	A(0), AD(3)
④ AB(5), DF(9), EG(3)	A(0), AD(3), DE(2)
⑤ AB(5), DF(9)	A(0), AD(3), DE(2), EG(3)

Goal state
Achieved

Step: ②:-

Minimum path Cost:-

$A \rightarrow D \rightarrow E \rightarrow G = 8$
 $3 + 2 + 3$

Ans

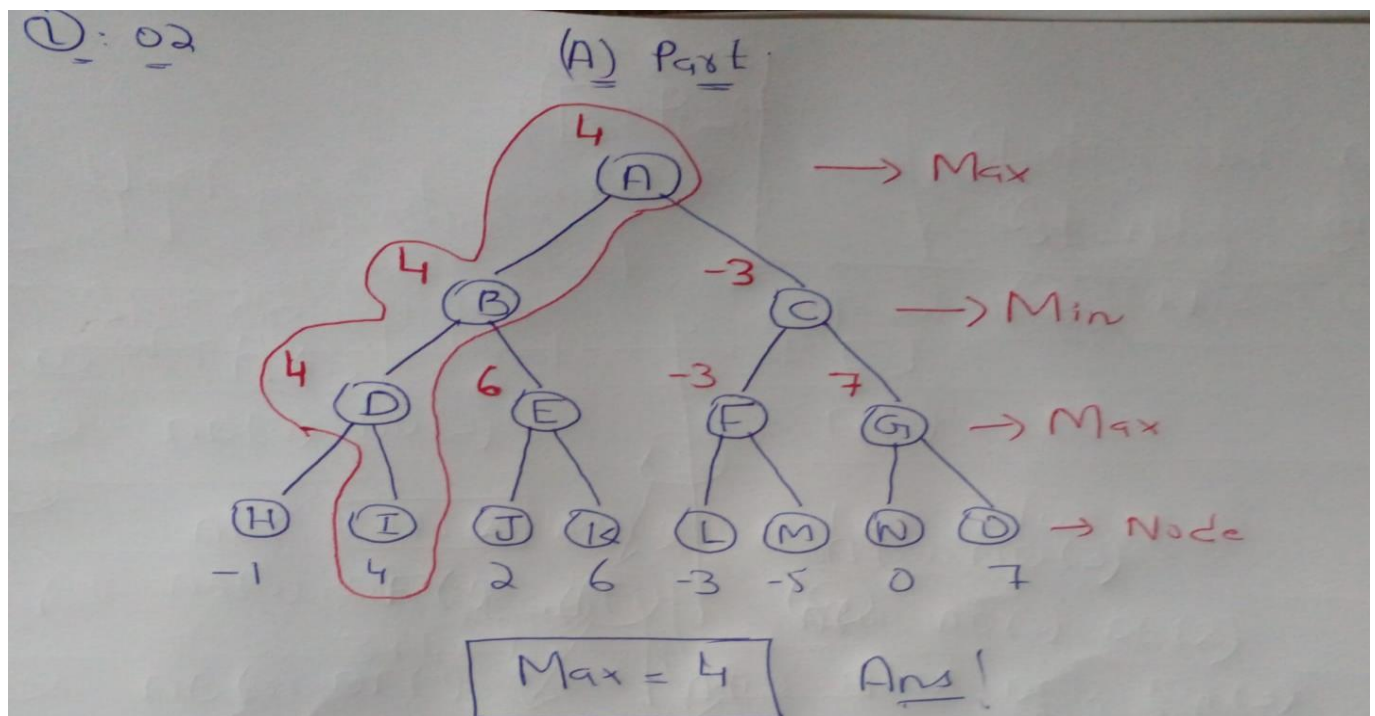
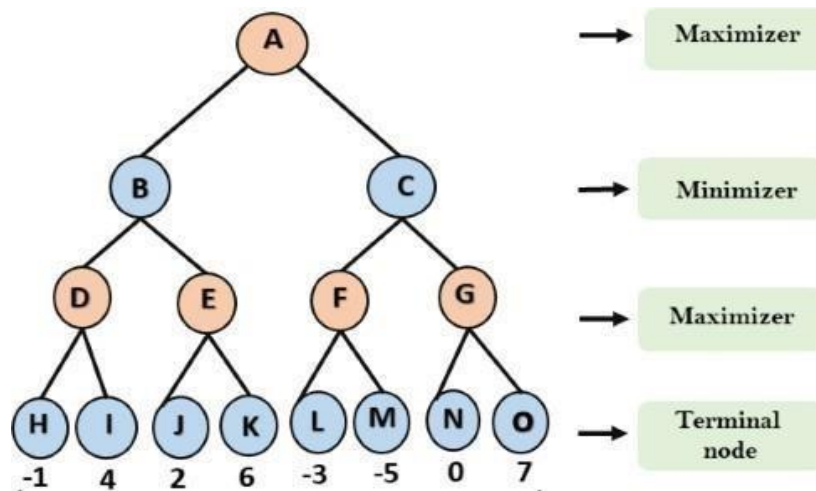


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Q.No.2 (CLO_2): (Cognitive Level C3, i.e., Applying)

(PLO_3: Problem Analysis)
(03+07 Marks)

- (a) **Consider** the Minimax Search Tree expanded by 2-players. **Apply** a Minimax Search on this tree in a depth-first manner searching from left to right. The values that the leaf-nodes will evaluate are given in the bottom most MAX layer.





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- (b) There is a need to satisfy equality i.e., $2a + b + 3c + 4d = 40$, genetic algorithm will be used to find the values of a , b , c , and d that satisfy the equation. The fitness function for this problem is:

$$f(x) = (2a + b + 3c + 4d) - 40$$

The objective is minimizing the value of fitness function $f(x)$. Since there are four variables in the equation, namely a , b , c , and d , we can compose the chromosome as follow: To speed up the computation, we can restrict that the values of variables a , b , c , and d are integers between 0 and 40.

Chromosome[1] = [a;b;c;d] = [02; 15; 18; 02]

Chromosome[2] = [a;b;c;d] = [10; 04; 12; 14]

Chromosome[3] = [a;b;c;d] = [14; 01; 10; 04]

Chromosome[4] = [a;b;c;d] = [11; 06; 15; 01]

- (i) **Calculate** the fitness for each chromosome and sort in ascending order.
- (ii) **Apply** uniform crossover at 2nd and 4th position on ranked 1st and 2nd.
- (iii) **Apply** one point crossover at middle point on ranked 3rd and 4th.
- (iv) **Calculate** the fitness of newly generated ranks also sort in ascending order and give your observation.
- (v) **Match** only the 1st rank fitness of step 4 is lower than the fitness of rank 1 in step 1.

ANS: 02 (B) PART

(i) Calculate fitness and sort ascending

Compute: $2a + b + 3c + 4d$

- **Chromosome 1:** $2(2) + 15 + 3(18) + 4(2) = 4 + 15 + 54 + 8 = 81 \rightarrow f(x) = |81 - 40| = 41$
- **Chromosome 2:** $2(10) + 4 + 3(12) + 4(14) = 20 + 4 + 36 + 56 = 116 \rightarrow f(x) = |116 - 40| = 76$
- **Chromosome 3:** $2(14) + 1 + 3(10) + 4(4) = 28 + 1 + 30 + 16 = 75 \rightarrow f(x) = |75 - 40| = 35$
- **Chromosome 4:** $2(11) + 6 + 3(15) + 4(1) = 22 + 6 + 45 + 4 = 77 \rightarrow f(x) = |77 - 40| = 37$

Sorted in ascending order:

Rank Chromosome # Genes [a, b, c, d] Fitness

1	3	[14,1,10,4]	35
2	4	[11,6,15,1]	37
3	1	[2,15,18,2]	41
4	2	[10,4,12,14]	76

(ii) Apply uniform crossover at 2nd and 4th position on ranked 1st and 2nd

- **Parents:** Chromosome 3 [14,1,10,4] and Chromosome 4 [11,6,15,1]
- **Positions:** 2nd, 4th



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Swap b and d $\rightarrow$ get:

- Offspring 1: [14, 6, 10, 1]
- Offspring 2: [11, 1, 15, 4]

(iii) Apply one-point crossover at middle point on ranked 3rd and 4th

- **Parents:** Chromosome 1 [2, 15, 18, 2] and Chromosome 2 [10, 4, 12, 14]
- **Middle point:** between b and c $\rightarrow$ split after b

Offspring:

- Offspring 3: [2, 15 | 12, 14]
- Offspring 4: [10, 4 | 18, 2]

Final:

- Offspring 3: [2, 15, 12, 14]
- Offspring 4: [10, 4, 18, 2]

(iv) Calculate fitness of newly generated chromosomes and sort

Compute:

- **Offspring 1:** $2(14)+6+3(10)+4(1)=28+6+30+4=68 \rightarrow f(x)=|68-40|=28$
- ****Offspring 2:**** $2(11)+1+3(15)+4(4)=22+1+45+16=84 \rightarrow f(x)=|84-40|=44$
- ****Offspring 3:**** $2(2)+15+3(12)+4(14)=4+15+36+56=111 \rightarrow f(x)=|111-40|=71$
- ****Offspring 4:**** $2(10)+4+3(18)+4(2)=20+4+54+8=86 \rightarrow f(x)=|86-40|=46$

Sorted in ascending order:

Rank Offspring Genes [a, b, c, d] Fitness

1	Offspring 1 [14, 6, 10, 1]	28
2	Offspring 2 [11, 1, 15, 4]	44
3	Offspring 4 [10, 4, 18, 2]	46
4	Offspring 3 [2, 15, 12, 14]	71

(v) Check if 1st rank fitness in step 4 is lower than 1st rank fitness in step 1

- Step 1 best fitness: 35
- Step 4 best fitness: 28

Yes, 1st rank fitness after crossover (28) is lower than before (35).

Observation

The genetic operations (uniform and one-point crossover) successfully generated an offspring with improved fitness, reducing the best solution from fitness 35 to 28, showing progress toward satisfying the equation constraint.



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